

Logical Reasoning Test

The Premises:

- If the Firewall is active, then the system is secure.
 $(F \rightarrow S)$
- If the system is secure, then data is encrypted
 $(S \rightarrow E)$
- The data base is corrupted or the Firewall is active
 $(C \vee F)$
- The data base is not corrupted.
 $(\neg C)$

Required:

Logically prove that data is encrypted (E).

Solution:

1. (Premise) $P \rightarrow S$
2. (Premise) $S \rightarrow E$
3. (Premise) $C \vee F$
4. (Premise) $\neg C$
5. $\neg C$ is true so C is false $\Rightarrow$ from premise 3
(false $\vee F$) $\Rightarrow F$
6. so we proof that F is true $\Rightarrow S$ is true
7. from premise 2 because S is true $\Rightarrow E$ is true
8. so E